

Positioning Composite: Methodology, Why It Is Slow, and How the Street Does It

Exact methodology of every positioning-labelled output · a quantified diagnosis of the slow crowding signal · a cited benchmark against CFTC/COT vendors, sell-side houses, the OFR and CTAs · implemented faster-normalization alternatives. Production default unchanged.

1. Exact methodology of the current outputs

Everything is built by `positioning.core` from three free sources: CFTC Traders-in-Financial-Futures (TFF, keyless), FRED yields, NY Fed primary-dealer cash. The pipeline, step by step (`file:function`):

1. **DV01 conversion** (`dv01.add_net_dv01`). Net contracts per contract (TU/FV/TY/TN/US/UB) $\times$ a per-contract DV01 (representative-CTD modified duration $\times$ price $\times$ 1bp) $\Rightarrow$ net DV01 in \$/bp, so tenors are additive in rate-risk units. This is the same “10-year-equivalent” idea the OFR Hedge Fund Monitor uses [7].
2. **Category duration** (`aggregate.category_duration`). Sum net DV01 across contracts within each trader category (leveraged funds, asset managers; dealer kept for the mirror).
3. **Feature panel, first z-score** (`aggregate.build_feature_panel`). For each category build `level_z` = **trailing 156-week (3y) z-score** of the net-DV01 *level*, plus a short-window flow `z` (`flow_z`, 52w), and three market features (curve, top-4 concentration, cash), each also 3y-z-scored (`normalize.rolling_zscore`, `window=156`, `min_periods=78`).
4. **PCA composite** (`aggregate.pca_factors`). PCA is fit on the 2006–2019 training slice and projected forward; PC1 (level positioning) is sign-aligned to the equal-weight average, then **re-standardized with a SECOND trailing 156-week z-score** to produce the headline *composite*. PC2 is the curve factor. An equal-weight fallback and a 156w rolling percentile accompany it.
5. **Point-in-time / leakage policy** (`score.run_score`). The composite is stamped on the CFTC *publication* date (report +3 bday COT lag); the predictive overlay (a small embargo-purged walk-forward logistic) and the conditional base-rate table read yields strictly after publication.
6. **Curve crowding** (`aggregate.curve_crowding`). Per category, `curve_duration` = front-end (TU+FV) minus long-end (US+UB) net DV01; 3y-z per category; the aggregate is the category mean, re-standardized (the raw mean is compressed by the -0.81 cross-category correlation). Positive = crowded steepener.

Read of “positioning” outputs: the composite z (crowding), its percentile, the stance (contrarian band on the composite), the DV01-by-trader $\times$ tenor heatmap (per-series 3y-z), the conditional base rates (historical P(yield up | crowded), descriptive), and the curve-crowding z . All normalization is the **3-year rolling z-score**; the composite applies it **twice**.

2. Why the composite is slow to flag over-crowding (quantified)

Two structural properties, both measured on the live data (`positioning.methodology_review`):

2.1 The 3-year window is a moving goalpost

A trailing 156-week mean and standard deviation are re-estimated each week from the last three years. When positioning stays elevated, the *mean itself drifts up into the crowd* and the *standard deviation inflates*, so the $z = (x - \mu)/\sigma$ is mechanically capped even as the raw book prints records. Figure 1 shows the buy-side net-DV01 stock breaking to new highs in 2025–26 while its own 3y mean chases it upward — the numerator never gets far ahead of the goalpost.

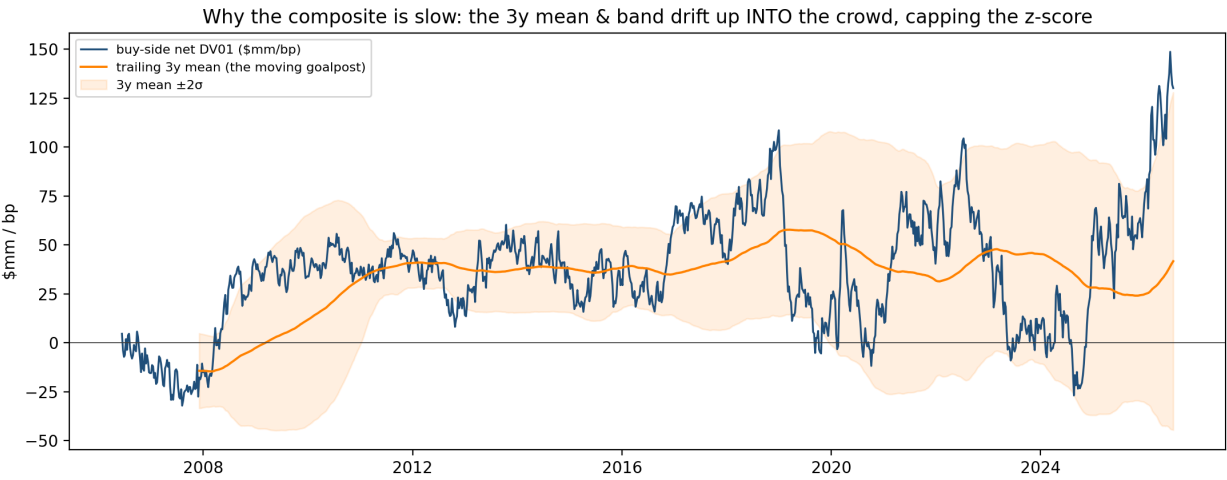


Figure 1: *
Fig 1. The 3y mean ($\pm 2\sigma$ band) drifts up into the rising position stock, capping the z-score — the mechanical source of the lag.

2.2 The double z-score adds a second lag

Features are 3y-z-scored, PCA-combined, then PC1 is 3y-z-scored *again*. Two passes of 3-year normalization compound the smoothing. Isolating it: a single-pass 3y-z of the same signal flags the current crowding roughly **7 months earlier** than the production (double-pass) composite.

2.3 Responsiveness: lead time to flag the current episode

Applying different normalizations to the *same* buy-side net-DV01 signal and asking when each first crossed “crowded” during the 2024–26 build-up:

Normalization (same signal)	first “crowded” flag	lead vs production	latest reading
Production composite (2× 3y-z + PCA)	2026-01-20	—	$z = +2.49$ (near cap)
Single-pass 3y-z	2025-07-01	+29 wks (~7m)	$z = +2.05$
COT-index, 3y (0–100)	2025-07-01	+29 wks	89.4
1-year z	2024-12-17	+57 wks (~13m)	$z = +1.39$
Net-% -of-OI, 1y z	2024-12-17	+57 wks	$z = +1.48$
EWMA z (half-life 26w)	2024-12-24	+56 wks	$z = +1.14$
COT-index, 1y (0–100)	2024-12-03	+59 wks	81.6
Percentile, 1y (0–100)	2024-06-11	+84 wks	86.5
Flow (13w change) 1y z	2024-11-19	+61 wks	$z = -0.64$

The lag is ~13 months. 1-year-based normalizations flagged this crowding in Nov–Dec 2024; the production composite waited until Jan 2026. The tell-tale signature of a laggy signal: the fast metrics have already *peaked and are rolling off* (1y-z +1.39, EWMA +1.14) while the slow composite is only now printing its high (+2.49) — it is a rear-view mirror. The flow metric is already *negative* (−0.64): the buy-side stopped adding months ago, which is precisely why the composite is topping out late.

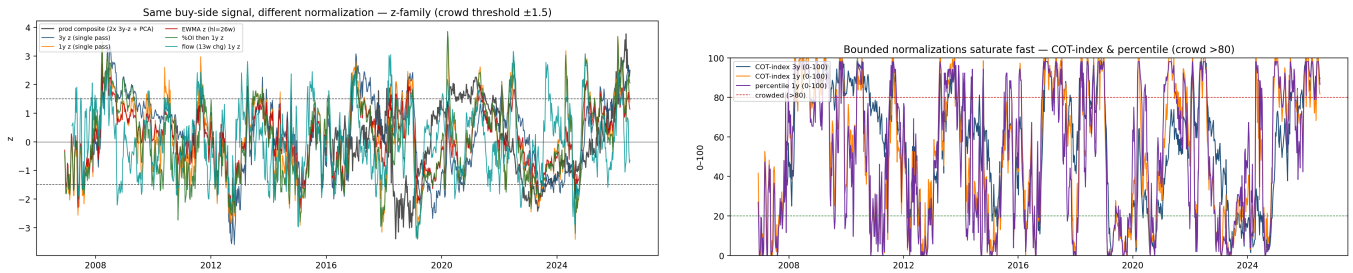


Figure 2: *
Fig 2. Same signal, different normalization. Left: z-family (crowd ± 1.5) — 1y/EWMA lead the 3y composite and turn earlier. Right: bounded COT-index / percentile (crowd > 80) — saturate fast but cannot rank “crowded vs extremely crowded”.

3. How Jefferies and the rest of the Street measure positioning (cited)

Direct answer on Jefferies: we found *no* public Jefferies CFTC-positioning framework — their rates names (Markowska, Simons) are macro/rates economists, not a quant positioning-index desk. The same is true of TD and RBC. Where banks publish a *rates* positioning gauge it is a **survey** (JPMorgan Treasury Client Survey net long–short % [14]; BofA Fund Manager Survey net OW/UW bonds), not a statistical z-score. Statistical positioning z-scores/percentiles that *are* documented (Goldman, Deutsche, Nomura) are equity/multi-asset-led and gate their base periods. So the honest comparison set is the public CFTC/COT methodology and the official-sector (OFR/Fed/TBAC) work.

Source	Normalization	Lookback	Bounded	Lvl/Flow	vs. our tool
Modal COT Index (Barchart, COT Un-chained) [1,2]	min-max $(x - \min)/(\max - \min)$	3y (also 1y)	0–100	Level	bounded, faster; no DV01/PCA
COT z-score (COTInsight) [3]	$(x - \mu)/\sigma$	52w standard	no	Level	<i>our normalization, but 1/3 our window</i>
Larry Williams (WillCo) [15]	min-max of net ÷ OI	26w	0–100	Level	OI-normalized, very fast
OFR Hedge Fund Monitor [7]	10y-equiv DV01 , raw \$	—	no	Level	<i>same DV01 idea we use</i> (best-practice match)
CFTC TFF (official) [4]	net, and net % of OI	—	—	Level	we use net; not %OI
US Treasury TBAC [5]	net % of Treasury market (ex-SOMA)	—	—	Level	growth-invariant; we don't do this
De Roan et al. (JF 2000) [10]	hedging pressure = net ÷ OI	—	—	Level	academic %OI convention
SG Trend Indicator (CTA) [13]	20/120-day MA cross, EWMA vol, vol-target	3m vol	—	Flow	positioning as vol-scaled flow, not a level
Spectra am/FX [-]	avg of %OI + 4w-change + skew-z, each rescaled ± 10	mixed	± 10	Both	bounded multi-source composite
Clarus / Litterman-Scheinkman [8]	PCA; PC2 = slope	—	no	Level	<i>we already use PCA (PC1/PC2)</i>
CME DV01-neutral 2s10s [9]	size legs so $DV01_1 = -DV01_2$	—	—	Level	our curve = front–long DV01 (same spirit)

Where our tool already leads the mode. The modal COT Index is a single-contract, level, min-max index with *no* risk-weighting. Our composite is *more* sophisticated on aggregation: it DV01-weights across tenors (matching the OFR's canonical 10y-equivalent method [7]) and uses PCA to de-collinearize (matching the Litterman-Scheinkman / Clarus factor approach [8]). **Where it lags the mode:** normalization. The Street's default lookback is 52 weeks (COTInsight explicitly warns 3y “reacts too slowly to structural shifts” [3]); many use bounded min-max; the sophisticated add *flow* (SG's trend model [13]; the academic change-premium [11]) and *%OI / market-size* normalization (CFTC [4], TBAC [5], De Roan [10]). We do the opposite: a 3-year window applied twice, on the level only.

Context (well sourced). The asset-manager-long / leveraged-fund-short split we report is the structure TBAC documents (AM synthetic-duration longs vs LF cash-futures basis shorts, $\sim 92\%$ R^2 since 2015) [5]; the LF Treasury-futures short reached $\sim \$0.7\text{--}1.1\text{tn}$ in 2023–24 per three Fed FEDS Notes and CFTC staff [6] — a reminder that a “crowded” TFF reading blends the basis trade with outright macro shorts.

4. Curve (steepener/flattener) positioning — methodology and alternatives

Current: $\text{curve_duration} = \text{front (TU+FV)} - \text{long (US+UB)} \text{ net DV01, per category, then the same 3y double-z.}$ This shares the CME DV01-spirit [9] and is a coarse version of PCA-PC2 [8], but inherits the same 3y lag *and* a level-vs-book ambiguity we already flagged: the current $z = +1.46$ (lev funds) reads “crowded steepener” only relative to a 3y window dominated by a prior flattener book; the **raw** front–long DV01 is $\approx$ flat ($-0.1 / -5.0$ \$mm/bp). Figure 3 separates the two dimensions: **book tilt** (raw, currently ≈ 0) vs **positioning momentum** (the z , steepener-direction). Better public templates: OFR 10y-equivalents [7], PCA PC2 loadings [8], and a 52w/EWMA or COT-index normalization instead of 3y.

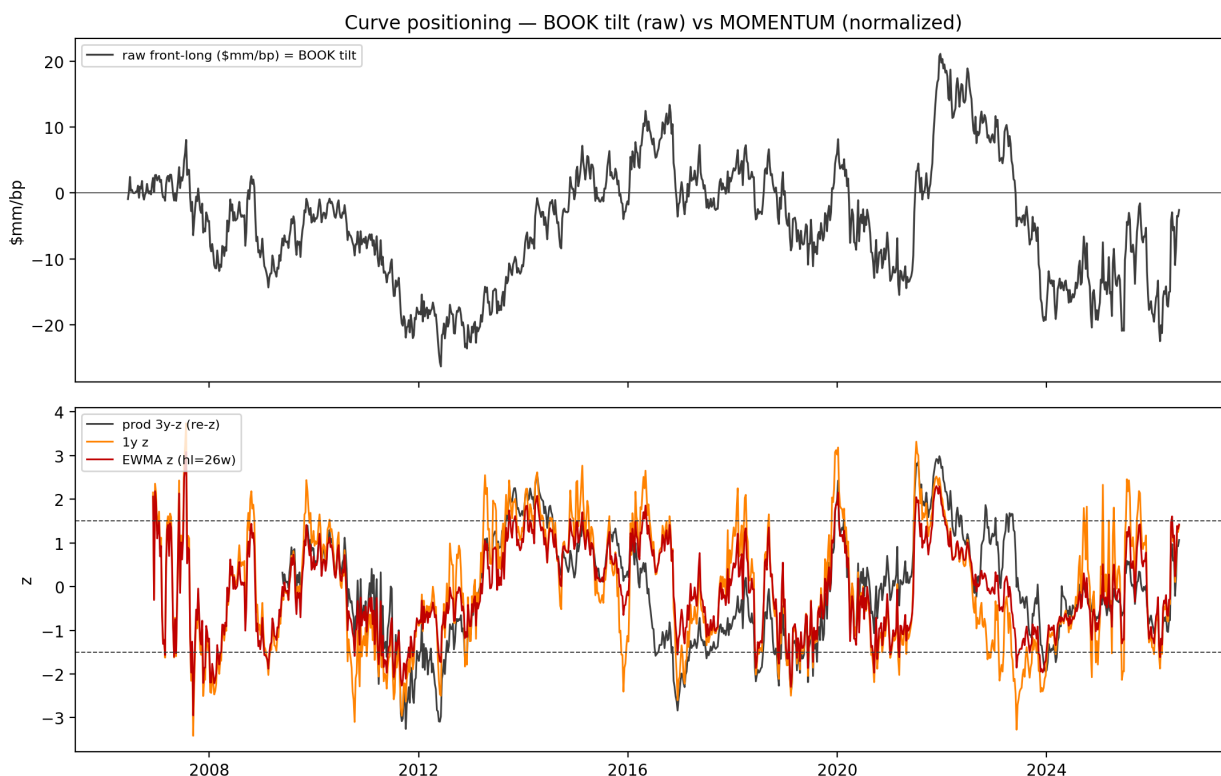


Figure 3: *

Fig 3. Curve positioning as two distinct things: raw book tilt (top, $\approx$ flat now) vs normalized momentum (bottom). The production 3y-z is the slowest line.

5. Recommendation

Keep what already matches best practice — **DV01 weighting and PCA** (OFR / Litterman-Scheinkman). Fix the normalization, which is where the tool is slow and where the Street is faster:

1. **Shorten & drop the double-z.** Standardize *once*, over ~ 52 weeks (or an **EWMA** mean/vol, half-life $\sim 26w$ [3,12]). On the current episode this alone would have flagged crowding ~ 7 – 13 months earlier.
2. **Normalize by open interest / market size** [4,5,10] so the score is crowding, not a bigger Treasury market.
3. **Surface FLOW alongside level** [11,13] — the 13-week change already turned negative here, a leading tell the level-z buried.
4. **Report a bounded COT-index / percentile as a secondary read** [1,2] for an easy 0–100 crowd gauge — accepting that it saturates (cannot rank crowded vs extremely crowded), so pair it with the (single-pass) z for magnitude.
5. **Curve: split book-tilt (raw DV01) from momentum (normalized)** and label them separately; consider PCA-PC2 [8].

Keep the 3-year z as a slow “structural context” band, not the headline. Net effect: a signal that reaches overcrowding on the Street’s timeline (weeks, not a year late) while retaining the tool’s risk-unit and factor sophistication. *The production default is unchanged in this note; the alternatives are computed side by side in results.positioning.method/.*

Sources. [1] COT Unchained, CoT Index (formula, 3y default, saturation): <https://cotunchained.com/en/Info/Commitments-of-Traders-CoT-Reports/CoT-Index>. [2] Barchart, COT Index (3y, 0–100): <https://www.barchart.com/education/technical-indicators/cot-index>. [3] COTInsight, COT z-score & window trade-off (52w; “3y too slow”): <https://cotinsight.com/blog/cot-z-score-explained>. [4] CFTC, TFF Explanatory Notes & % of open interest: <https://www.cftc.gov/MarketReports/CommitmentsofTraders/ExplanatoryNotes/index.htm>. [5] US Treasury TBAC, Q1-2024 (AM-long/LF-short, market-size normalization): <https://home.treasury.gov/system/files/221/TBACCharge1Q12024.pdf>. [6] Federal Reserve FEDS Note, hedge-fund Treasury futures/repo (30 Aug 2023): <https://www.federalreserve.gov/econres/notes/feds-notes/recent-developments-in-hedge-funds-treasury-futures-and-repo-positions-20230830.html>. [7] OFR Hedge Fund Monitor, 10-year-equivalent DV01 method: <https://www.financialresearch.gov/hedge-fund-monitor/datasets/tff/>. [8] Clarus FT, PCA of the swap curve (PC2 = slope); Litterman & Scheinkman (1991): <https://www.clarusft.com/principal-component-analysis-of-the-swap-curve-an-intro>. [9] CME, DV01-neutral yield-curve spreads: <https://www.cmegroup.com/education/files/yield-curve-spread-trades.pdf>. [10] De Roan, Nijman & Veld, hedging pressure = net÷OI, J. Finance (2000): <https://onlinelibrary.wiley.com/doi/10.1111/0022-1082.00253>. [11]

Kang, Rouwenhorst & Tang, position *changes* vs levels: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2449315. [12] EWMA responsiveness (RiskMetrics λ ; Boyd et al.): <https://web.stanford.edu/~boyd/papers/pdf/ewmm.pdf>. [13] SG Trend Indicator methodology (CTA replica): https://wholesale.banking.societegenerale.com/fileadmin/indices_feeds/SG_Trend_Indicator_Methodology_Summary.pdf. [14] JPMorgan Treasury Client Survey (secondary paraphrase): <https://uk.investing.com/news/economy-news/treasury-note-investor-sentiment>. [15] Larry Williams / WillCo min-max lineage (net÷OI, 26w): <https://cotunchained.com/en/Info/Commitments-of-Traders-CoT-Reports/Williams-Commercial-Index>. Confidence: [1–13] primary/vendor-methodology or journal; [14] and all bank *rates* indices are secondary paraphrase (no house publishes a rates z-score window). No Jefferies/TD/RBC public CFTC framework was found. The Williams→COT-Index link is algebraic lineage, not a sourced derivation.